THE BAY RIG

A 10-Foot Inflatable Fishing Boat for San Diego Bay — A DIY Build and Six-Month Study

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A 10-foot inflatable, a 6 HP outboard, and transom wheels — a six-month study in getting on San Diego Bay to fish.

Jonas Bonnett · November 2026

How to read this paper

This paper is a factual account of a personal DIY project: a 10-foot inflatable boat, a 6 HP outboard, and transom wheels, built up over six months so its owner could fish San Diego Bay. Everything in it comes from one of two places: what Jonas Bonnett gave (his words, his photo), or a published specification or an official regulation that is cited with its source. This is a factual account of one person's build — not a manual for anyone else's boat, and not legal advice. Boating regulations change; check the California Division of Boating and Waterways before relying on any rule stated here.

Where a fact is not yet known, the paper says so plainly with a [SME VERIFY] flag instead of guessing. A figure that comes from a manufacturer's listing is labeled as published, not measured. Values marked [PROPOSED] are standard choices for this class, filled in for approval — not measurements from this rig.

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1. The Problem

San Diego Bay is a big, productive inshore fishery, and fishing it from shore means fishing the piers and the shoreline. A boat opens the rest of it — the channels, the flats, the structure, the water you cannot reach on foot. But a conventional boat brings a trailer, a tow vehicle, storage, a slip or a launch-ramp routine, and thousands of dollars before a line ever gets wet.

The problem this project set out to solve was simple: get on San Diego Bay to fish, without a trailer, without a slip, and without a big boat. The constraints that followed from that were just as simple. The rig had to be portable — transportable without a trailer. It had to be launchable by one person. It had to be affordable. It had to carry one person, fishing gear, and a motor. And it had to be legal on California waterways, with a licensed operator.

The answer the project arrived at was a 10-foot inflatable boat, a 6 HP four-stroke outboard, and a set of transom wheels. The inflatable packs down for transport. The wheels let one person roll the rig from the car to the water. The 6 HP outboard gives it range under power across the bay. The oars stay aboard as backup. Nothing about the rig is exotic; the point of the project was to put ordinary parts together the right way, after studying what the right way was.

2. The Study

This was a six-month project, and most of the six months was study, not building. Two things were studied before the rig was complete: the law, and the motor.

2.1 The license

Jonas Bonnett took his boating license to understand the rules and regulations. In California, that license is the California Boater Card. The law behind it is Senate Bill 941, signed in 2014, which prohibits operating a motorized vessel in California without a valid operator card issued by the Division of Boating and Waterways. The requirement was phased in by age group starting in 2018, and as of January 1, 2025 it applies to every operator of a motorized vessel on California waterways, regardless of age.

Getting the card means completing a boating safety course approved by the Division of Boating and Waterways and by NASBLA, the National Association of State Boating Law Administrators, passing the exam at the end of the course, and applying for the card. The card costs $10 and is valid for life. The course covers the rules of the road on the water, safety equipment, and the operator's responsibilities — the knowledge the card is meant to prove.

That is why the license came first in this project. The rig has a motor, so the card is required to operate it, and the study of the rules was part of the build, not an afterthought.

2.2 Motor setups for different purposes

The second part of the study was motor setups: different motors for different purposes. A 10-foot inflatable can take several setups, and each one fits a different kind of day on the water. The table below is background — the usual setups for a boat this size, with the purpose each one serves. It is not a record of the study itself: which setups Jonas Bonnett actually compared, and what he concluded about each, is an open question at the end of this paper [SME VERIFY].

Motor setups for a 10-foot inflatable — background, not study notes

Setup	What it is for	Trade-off
Oars only	Silent, close-in fishing; no fuel, no motor to maintain	Slow, short range, hard work against wind and current
Electric trolling motor	Quiet, precise boat control while fishing	Limited speed and range; battery weight and charging
Small 2-stroke outboard	Light and simple; easy to carry and mount	Louder, smokier, fuel mixed with oil
4-stroke 6 HP outboard — SELECTED	Bay range under power; quiet, fuel-efficient, no oil mixing	Heavier than a 2-stroke; small built-in tank
Larger outboard, 9.9–15 HP	Speed and power for bigger water	Heavier, more fuel, more than a 10-foot inflatable's transom and the bay's no-wake zones call for

The choice that came out of the study was the 4-stroke 6 HP. For fishing San Diego Bay — protected water, no-wake zones, short runs between spots — 6 HP is the right size: enough power to move the boat and hold it against wind and current, quiet enough not to spook fish, and light enough (about 20 kg) to carry and clamp to the transom by hand. A larger motor would add weight and fuel burn without adding anything the bay asks for.

3. The Rig

The rig has three parts: the boat, the motor, and the wheels. Each one is ordinary. Together they make a boat that one person can take from the car to the bay.

3.1 The boat

A 10-foot inflatable. The tubes are grey with a red and blue rub strake, and the boat carries a bench seat. The brand and model are unknown — deliberately unnamed in this paper, so the story is about the build, not an advertisement for the maker. The tube material, floor type, and rated capacity are filled in below as standard choices for a 10-foot inflatable of this class, pending his approval:

- Tube material: PVC, 1100 denier / 0.9 mm [PROPOSED]
- Floor: high-pressure air deck, drop-stitch [PROPOSED]
- Rated capacity: 4 persons, 550 kg maximum load [PROPOSED]

Ten feet is the right size for the job. It is big enough to be stable with a person, a motor, and gear aboard, and small enough to transport without a trailer and to launch by hand. It is a bay boat, not an ocean boat — that limit is stated plainly in Chapter 5.

3.2 The motor

A 6 HP four-stroke outboard, tiller-steered, hand-pull start. The specifications below are published figures for the motor on this rig, taken from retailer listings — not measurements from this rig. Where listings disagree, the range is shown.

6 HP four-stroke outboard — published specifications

Item	Published figure
Max output	6.0 HP (3.75 kW)
Engine	Single cylinder, 4-stroke, air-cooled
Displacement	140 cc
Max speed	4,500–5,500 rpm
Gear ratio	1:1.5 (16/24)
Fuel tank (built-in)	1.2 L
Fuel consumption (published)	about 1 L/h
Weight (net, published)	about 20 kg (listings give 19.7–21 kg)
Control	Tiller, manual tilt, hand-pull start
Clutch	Centrifugal
Ignition	Transistor non-contact electronic (TCI)
Propeller range (published)	3–7 1/2 in
Shaft	Short shaft — 15 in (381 mm), the standard short-shaft length [PROPOSED]
Speed (published range)	about 4–15 km/h, depending on environment, water, and hull load — not measured on this rig

Two things about the motor matter for this rig. First, it is air-cooled, so there is no water pump or impeller to maintain — one less thing to fail on a small boat. Second, it is a four-stroke, so it runs on straight gasoline with no oil mixing, it is quieter than a two-stroke, and it sips fuel: about a litre an hour at the published figure, which makes the small 1.2-litre built-in tank workable for short bay runs.

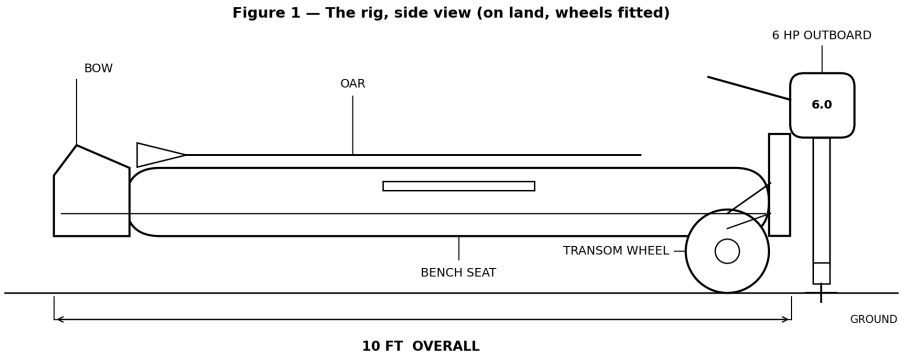


Figure 1 — The rig, side view. A 10-foot inflatable with the 6 HP outboard on the transom and a transom wheel fitted for transport. Dimensions are the rig's own: 10 ft overall, motor about 20 kg. Not to scale.

3.3 The wheels

Transom wheels, installed for portable transport. The wheels were not a bolt-on afterthought: Jonas Bonnett researched transom wheels for this inflatable, drilled holes in the transom for the mounting brackets, and installed the wheels himself. They bolt to the transom and let one person roll the inflated boat from the car to the water — no trailer, no second pair of hands. The wheel size and type are filled in as the standard choice for this class, pending approval: 10 in × 3 in pneumatic wheels on stainless-steel brackets [PROPOSED]. The photo shows one red-rimmed wheel under the starboard tube, consistent with that size.

The wheels are a transport fitting, not a running fitting: they carry the boat on land, and they come up or off before the boat runs under power. The launch sequence in Chapter 4 covers that step.

3.4 How it fits together

The motor clamps to the transom. The wheels bolt to the transom on mounting brackets, through holes drilled for the brackets. The oar ships on the port tube. The kill-switch lanyard — the red coiled cord visible in the photo — clips to the operator, so the motor stops if the operator goes overboard. The bench seat carries the operator, and the gear stows inside the tubes. Nothing is permanently fixed except the wheels; the motor comes off the transom for transport and storage.

3.5 The leak, and the fix

After the wheels were installed, Jonas Bonnett noticed a small water leak from the engine. He sealed the drilled holes with boating epoxy, bought at Home Depot and Ace Hardware. The repair was tested in the water, and it passed — the seal held. Drilling the transom is the one irreversible step in this build — every other part of the rig comes off — so the seal is what keeps the transom sound.

3.6 What it cost

The whole setup cost $1,210. The breakdown is what was paid:

What the rig cost

Item	Cost
Inflatable boat	$450
6 HP outboard engine	$500
Transom wheels	$60
Miscellaneous	$200
Total	$1,210

That is the affordability constraint from Chapter 1, met in dollars: a fishable bay boat for $1,210, with no trailer, no tow vehicle, and no slip on top of it.

4. Launch and Operation

4.1 Launch sequence

The sequence below is the standard launch for a wheeled inflatable of this size. It is written here as the working sequence for this rig — Jonas Bonnett has not confirmed the order, and it should be corrected to his actual routine [SME VERIFY].

1. Inflate the tubes and check them firm, and check the transom, the seat, and the floor are secure.
2. Fit the transom wheels.
3. Mount the motor on the transom and tighten the clamps.
4. Check fuel in the built-in tank and engine oil.
5. Load gear, and put a USCG-approved life jacket aboard for each person.
6. Roll the rig from the car to the water on its wheels.
7. Float the boat, then raise or remove the wheels.
8. Clip on the kill-switch lanyard.
9. Start the motor in the water — never run it dry on land.
10. Idle clear of the ramp, then run.

4.2 Pre-launch checklist

[ ] California Boater Card aboard.

[ ] USCG-approved life jacket aboard for each person.

[ ] Kill-switch lanyard aboard and clipped on.

[ ] Fuel checked; oil checked.

[ ] Motor clamps tight.

[ ] Oar aboard.

[ ] Transom wheels raised or removed before running.

[ ] Weather and wind checked.

[ ] Someone ashore knows where you are going and when you expect to be back.

4.3 On the water

The rig is a bay boat. It runs the bay under power, idles into fishing spots, and holds position while fishing. The oars are the backup if the motor stops. The photo below is the rig at the ramp, ready to launch: motor on the transom, wheel fitted, oar shipped, gear aboard.



Figure 2 — The rig at the boat ramp, stern view. 6 HP outboard on the transom, tiller steering, red kill-switch lanyard, oar on the port tube, red transom wheel under the starboard tube. Photograph: Jonas Bonnett.

5. What It Does and What Comes Next

5.1 The result

Six months of study and a simple rig produced a boat that does what the project asked for: one person can take it from the car to San Diego Bay and fish, with no trailer, no slip, and no big boat. The operator is licensed, the rig is legal, and the motor is the right size for the water it will run. The transom wheels are the detail that makes the whole thing work — without them, a 10-foot inflatable with a 20-kg motor is a two-person carry; with them, it is a one-person roll.

5.2 Honest limits

The limits are part of the result, not a footnote to it. Ten feet and 6 HP is a bay boat. It is not an ocean boat, and it should not be taken outside the bay. The built-in tank holds 1.2 litres, which at the published consumption of about a litre an hour is roughly an hour of running — range is a real limit, and the fuel plan for a day on the bay is an open question [SME VERIFY]. The published speed range of about 4–15 km/h is the manufacturer's, not a measurement from this rig, and it depends on load, water, and wind. The boat's rated capacity is a proposed figure — 4 persons and 550 kg, the standard for the 10-foot class — pending his confirmation [PROPOSED]. Until it is confirmed, the safe load is one person plus gear — which is how the rig is used.

5.3 What comes next

Three things come next. First, the open questions in Appendix C — the boat's capacity, the motor's shaft length, the launch ramp, the fuel plan, and the fishing results. Second, a first fishing report: what was caught, where, and how the rig behaved. Third, the connection to the fishing work already published: the Shore Fishing Starter Guide for San Diego, the Knots guide, and the California Lobster and Crab Guide are all shore-based. This boat extends that work from the shore onto the bay, and a bay-fishing companion to those guides is the natural next product — when the fishing results are in.

Appendix A. Spec Sheet

A.1 The rig

The rig

Item	Figure
Boat	10 ft inflatable — brand and model unknown (deliberately unnamed)
Tube material	PVC, 1100 denier / 0.9 mm [PROPOSED]
Floor type	High-pressure air deck (drop-stitch) [PROPOSED]
Rated capacity	4 persons, 550 kg max load [PROPOSED]
Motor	6 HP, 4-stroke, single cylinder, air-cooled
Motor weight (published)	about 20 kg (19.7–21 kg across listings)
Motor shaft	Short shaft — 15 in (381 mm) [PROPOSED]
Fuel tank (built-in)	1.2 L (published)
Fuel consumption (published)	about 1 L/h
Transom wheels	10 in × 3 in pneumatic, stainless-steel brackets [PROPOSED]
Wheel mounting	Holes drilled in the transom for the mounting brackets; sealed with boating epoxy (Home Depot / Ace Hardware)
Oar	One, shipped on the port tube

A.2 Sources

Motor specifications are the published figures for the motor on this rig — a 6 HP four-stroke air-cooled outboard — taken from retailer listings. Where listings disagree — notably net weight, 19.7–21 kg — the range is shown rather than a single figure. No motor figure in this paper was measured on this rig.

Appendix B. Rules Checklist

These are the rules that apply to this rig, from official sources. Regulations change — verify with the California Division of Boating and Waterways before relying on them.

- California Boater Card: required for every operator of a motorized vessel on California waterways since January 1, 2025 (Senate Bill 941; Division of Boating and Waterways). Obtained by completing an approved boating safety course and passing the exam; $10, valid for life. Source: Division of Boating and Waterways, Boating Education and Outreach report (November 2025); CaliforniaBoaterCard.com.
- Life jackets: California law requires a USCG-approved life jacket aboard for each person. Every child under 13 must wear a USCG-approved life jacket at all times while on a moving vessel (Harbors and Navigation Code section 658.3). For a boat under 16 feet, one USCG-approved wearable life jacket per person aboard is the requirement. Source: California State Parks, Boating laws, rules and license requirements (parks.ca.gov).
- Age: a person must be 16 or older and hold a California Boater Card to operate a vessel powered by a motor of 15 HP or more. This rig is 6 HP, so that threshold does not apply — but the Boater Card requirement does. Source: California State Parks, Boating laws, rules and license requirements.
- This paper is not legal advice. It states the rules the project studied; it does not interpret them for anyone else's boat.

Appendix C. Open Questions

These are the facts only Jonas Bonnett can supply. The paper is complete without them; answering them makes it exact. Questions 1–3 carry proposed values in the paper, marked [PROPOSED]; the brand and model are deliberately unnamed.

1. Confirm or correct the proposed tube material, floor type, and rated capacity.
2. Motor shaft length, and whether the motor was bought new or used.
3. Transom wheel size, type, and how they are raised or removed before running.
4. The launch ramp used in San Diego Bay.
5. The fuel plan for a day on the bay — spare fuel carried, and how much.
6. Which motor setups were actually compared, and what was concluded about each.
7. The dates of the six months.
8. Fishing results: what was caught, where, and how the rig behaved.
9. Whether the oar, the kill-switch lanyard, and the pre-launch checklist match his actual routine, and the correct order of the launch sequence in Chapter 4.